



Optimizing stand-alone microgrids with lagrange multiplier technique: A cost-effective and sustainable solution for rural electrification

Rasha Elazab^{*}, Ahmed Mamdouh Ewais, Maged Ahmed Abu-Adma

Electrical Power and Machines Department, Faculty of Engineering, Helwan University, Cairo, Egypt

ARTICLE INFO

Keywords:

Lagrange multiplier optimization
Stand-alone microgrids
Renewable energy systems
Hybrid microgrid planning
Cost analysis

ABSTRACT

This paper addresses the challenge of optimally sizing and planning stand-alone microgrids in remote areas, focusing on generating sources using a novel algorithm based on the Lagrange multiplier optimization technique. The study centers on the Gulf of Aqaba, Egypt, where five configurations of PV and wind micro plants with diesel generation are evaluated. Detailed cost analysis includes wind turbines, PV modules, and diesel generators. Motivated by the need for cost-effective, reliable, and environmentally friendly microgrids, the proposed algorithm is benchmarked against the widely used Hybrid Optimization Model for Energy Renewable HOMER software. Key criteria for analysis include economic benefits, environmental impacts, and system reliability. Results highlight the superior performance of the proposed optimization technique. It achieves up to 22 % lower net present costs (NPC) and up to 13 % lower cost of electricity (COE) compared to HOMER. Additionally, the algorithm demonstrates significant environmental benefits, with emissions reductions of up to 25 % for carbon dioxide and substantial decreases in carbon monoxide and nitrogen oxides. Reliability is also enhanced, with the proposed schemes showing higher excess energy and renewable energy contributions. The study justifies the use of the Lagrange multiplier optimization technique as a superior approach for planning and sizing stand-alone microgrids, offering significant economic and environmental advantages. This work provides a comprehensive framework for developing sustainable energy solutions in isolated regions.

1. Introduction

Conventional energy sources increase greenhouse gas emissions, threatening global climate stability. Additionally, the rising cost of fossil fuels and their unavailability in certain regions further highlight the need for alternative energy sources. Consequently, clean alternatives such as wind and solar power have become more attractive and widely adopted for substituting fossil fuels in electrical power generation. Microgrids (MGs), which leverage these renewable sources, aim to provide high-quality, reliable energy at lower costs and with reduced carbon emissions. MGs can operate either interconnected with the utility grid or as isolated systems [1,2].

Recent studies have explored various optimization techniques for MG planning and sizing [3]. For instance, the Multi-Grey Wolf Optimizer (GWO) was used to optimize the sizing of different components in hybrid microgrid systems to meet the electrical load of small villages in Malaysia [4]. The Multi-Objective Wind-Driven Optimization (MOWDO) method was employed to optimize operation costs, pollution emissions, and availability with/without hybrid demand response

programs and incline block tariff [5]. Multi-Objective Genetic Algorithm (MOGA) focused on optimizing the sizing of multi-source PV/Wind with hybrid energy storage systems to balance the total cost of electricity and the loss of power supply probability [6]. The Multi-Objective Particle Swarm Optimization (MOPSO) provided a framework for designing and optimizing microgrids in rural areas with scattered energy resources to meet maximum load demands under changing weather conditions [7].

Additionally, the Multi-gravitational Search Algorithm (MOGSA) proposed multi-objective optimization methods for the sizing of a hybrid AC/DC microgrid system integrating renewable energy sources [8]. Other notable techniques include the African vulture optimization approach (AVOA), which achieved good economic results for photovoltaic panels, wind turbines, diesel generators, and batteries [9]. Despite their advantages, these sophisticated techniques have several disadvantages. GWO can be computationally intensive and may not always provide the optimal solution due to its dependence on specific problem characteristics and constraints. MOWDO lacks a strong theoretical foundation, making it difficult to predict its performance in different scenarios. MOGA's mechanisms may limit solution diversity,

^{*} Corresponding author.

E-mail address: r_m_elazab@h-eng.helwan.edu.eg (R. Elazab).

<https://doi.org/10.1016/j.fraope.2024.100199>

Received 27 June 2024; Received in revised form 29 November 2024; Accepted 10 December 2024

Available online 11 December 2024

2773-1863/© 2024 The Author(s). Published by Elsevier Inc. on behalf of The Franklin Institute. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

while MOPSO struggles with problems having many objectives. MOGSA may face challenges in balancing exploration and exploitation, risking premature convergence or insufficient search space exploration.

The Parallel Particle Swarm Optimization (PPSO), Parallel Vortex Search Algorithm (PVSA), and Parallel Ant Lion Optimizer (PALO) have been used to optimize the operation of batteries in standalone and grid-connected DC microgrids, significantly reducing energy costs and CO₂ emissions [10]. However, these methods suffer from increased communication overhead, which can become a bottleneck as the number of processors or threads increases. The Hybrid Particle Swarm Optimizer-Grasshopper Optimization Algorithm (GOA) aims to improve the eco-techno-economic performance of microgrids with supercapacitor-integrated batteries, but its solutions may not be robust to changes in weather patterns or load demand [11]. The Lightning Search Algorithm (LSA) is designed to minimize the energy production cost of diesel generators and the costs of power exchange within microgrids, but it has limited theoretical analysis and guarantees regarding its convergence properties, making it harder to predict its performance in different scenarios [12].

The Salp Swarm Algorithm (SSA) optimizes the configuration of integrated hybrid renewable energy systems, yet it is highly sensitive to the selection and tuning of its parameters, which can result in suboptimal convergence [13]. The Social Spider Optimizer (SSO) focuses on minimizing the cost of energy by determining the optimal number of PV panels, wind turbines, and battery autonomy days, but its accuracy depends heavily on the accuracy of input data such as weather and load demand [14]. The Pelican Optimization Algorithm (POA) aims to reduce overall operating costs, including those of conventional generator fuel and power transactions, but it relies on the assumption that customers will respond predictably to incentive-based demand response programs, which may not always be the case [15]. The Harris Hawks Optimization (HHO) algorithm applies Benson's scalarization method to improve multi-objective optimization, but its results are based on a single case study, limiting their generalizability [16]. Lastly, the grasshopper optimization algorithm proposes a rule-based energy management scheme to minimize energy costs and the deficiency of power supply probability, but as a stochastic algorithm, its convergence to the optimal solution is not guaranteed, and results may vary depending on initial conditions and random seed used [17].

The Lagrange multiplier optimization method, a technique with decades of proven efficacy in power system economic analysis, is particularly advantageous for planning microgrids in rural, off-grid, low-income areas. This well-established method offers a straightforward and efficient approach to optimizing the sizing and operation of hybrid renewable energy systems, including photovoltaic panels, wind turbines, diesel generators, and batteries. Unlike more sophisticated optimization techniques, which can be computationally intensive and complex to implement, the Lagrange method is simple, easy to code, and requires fewer computational resources, making it ideal for regions with limited access to advanced technological tools and expertise. Its application ensures reliable, cost-effective, and environmentally friendly energy solutions by efficiently utilizing available natural resources and meeting load demands. This method's flexibility and maturity make it an excellent choice for developing sustainable microgrid systems that can provide high-quality, continuous power to remote communities, thereby supporting their development and improving their quality of life. The Lagrange multiplier's robust track record in enhancing planning processes highlights its significant potential to address the unique challenges faced by rural electrification projects.

This research presents a comprehensive study of stand-alone microgrids, focusing on the optimal sizing and planning of generating sources using a novel algorithm based on the Lagrange multiplier optimization technique. The study centers on the Gulf of Aqaba, Egypt, where five configurations of PV and wind micro plants with diesel generation are evaluated. Detailed cost analysis includes wind turbines, PV modules, batteries, fuel cells, and diesel generators. Key criteria for

analysis include economic benefits, environmental impacts, and system reliability. The results demonstrate the superior performance of the proposed optimization technique. It achieves lower net present costs and lower cost of electricity compared to HOMER. Additionally, the algorithm shows significant environmental benefits, reducing carbon dioxide emissions along with substantial decreases in carbon monoxide and nitrogen oxides. Reliability is also enhanced, with the proposed schemes exhibiting higher excess energy and renewable energy contributions.

The paper is organized as follows: Section 2 provides an overview of the technical background needed for MG's technical models. Cost analysis is presented in Section 3. Section 4 discusses the proposed optimization algorithm and its implementation. Section 5 compares the results obtained using the proposed technique with those from the HOMER software, focusing on economic, environmental, and reliability metrics. Section 6 concludes the paper with a discussion of the findings.

2. Technical background

MGs usually consist of both traditional and renewable micro plants. Traditional sources, such as diesel generators, ensure the reliability of MGs since they can provide power regardless of weather conditions. Conversely, renewable sources are essential for MGs due to their clean nature and suitability for distributed energy generation. PV and wind micro plants are common technologies in contemporary MGs.

Energy storage devices play a crucial role in modern grids. They store excess energy produced by renewable sources within MGs, compensating for any power shortages. Various types of energy storage are commercially available today. The following sections discuss the main technical and economic features of the key MG components [18].

2.1. Wind turbines

The output power of wind plants varies with instantaneous wind speed, posing a significant challenge in planning and operating processes. Wind power output is typically described by a power curve, which shows the steady-state electrical power output as a function of wind speed at the hub height. Power output increases cubically with wind speed until the rated wind speed is reached. Wind plant power can be modeled using the power curve equation [19]:

$$PWT(t) = \begin{cases} a \cdot V^3(t) - b \cdot P_R & V_{ci} < V < V_r \\ P_R & V_r < V < V_{co} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

Where: $a = PR / (V_r^3 - V_{ci}^3)$, $b = V_{ci}^3 / (V_r^3 - V_{ci}^3)$. V_{ci} , V_r , and V_{co} are cut-in, rated, and cut-out wind speeds, respectively.

2.2. Photovoltaic modules PV

PV modules allow for clean, quiet, and reliable electricity generation. PV power generation systems are environmentally friendly as they produce no pollutants or greenhouse gases during operation. The power output of PV modules depends primarily on global solar radiation, which varies daily and seasonally.

The power generated by the PV array is calculated as [20]:

$$P_{pv} = Y_{pv} F_{pv} (G_T | G_{T,STC}) [1 + \alpha_p (T_c - T_{c,STC})] \quad (2)$$

Where: Y_{pv} is the rated capacity of the PV array, meaning its power output under standard test conditions [kW], F_{pv} is PV derating factor [%], G_T is the solar radiation incident on the PV array in the current time step [kW/m²], $G_{T,STC}$ is the incident radiation at standard test conditions [1 kW/m²], α_p is the temperature coefficient of power [%/°C], T_c is the PV cell temperature in the current time step [°C], $T_{c,STC}$ is the PV cell temperature under standard test conditions.

2.3. Diesel generators

Diesel generators are used in locations without grid connections or as emergency power supplies. They are also used in more complex applications such as short-term operating reserves to support national grids during sudden power demand spikes or power plant losses. Factors to consider when choosing a diesel generator include load type, load curve, power factor, and starting method. Diesel power output is related to fuel consumption and can be expressed as [20]:

$$C_i = \alpha + \beta_i P_i + \lambda_i P_i^2 \text{ \$/hr(3) } \text{ Where: } C_i, P_i \text{ is the diesel fuel cost and diesel power, respectively.}$$

2.4. Batteries

Batteries are widely used for various applications, including portable electronics, power tools, hybrid vehicles, and as energy backups. They are becoming more efficient, cost-effective, and popular. Batteries are sensitive to maintenance and proper operation but offer advantages such as rapid response, small size, no emissions, and quiet operation, making them ideal for smoothing renewable energy fluctuations. Common battery types include Lead-Acid, Li-ion, and Nickel-based batteries, each with distinct features. Lead-Acid batteries, used in this study, have a high efficiency of over 87 % and a low self-discharge rate [18].

The state of charge (SOC) of batteries indicates the available stored energy and is evaluated hourly based on the MG's load profile, wind speed, and solar radiation. SOC is defined as [21]:

$$SOC(t) = SOC(t-1) \cdot (1 - \delta_{bat}(t)) + \frac{Pb(t)}{V_{bus}} \cdot \eta_{bat} \cdot \Delta t \quad (4)$$

with the constraint:

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (5)$$

Where : $SOC(t)$, $SOC(t-1)$ are state of charge at a time t , $t-1$, respectively, $\delta_{bat}(t)$, is the hourly battery self-discharge rate, $Pb(t)$ is the energy stored in the battery at hour t in kWh, V_{bus} is the dc bus voltage, η_{bat} is the battery round trip efficiency, and SOC_{min} , SOC_{max} are the minimum and maximum permissible SOC, respectively.

2.5. Hydrogen fuel cells

Hydrogen fuel cells, similar to batteries, use electricity to produce hydrogen during charging. During discharge, hydrogen feeds the fuel cell to generate electricity. Hydrogen is produced from water via electrolysis. Unlike batteries, fuel cells are not closed systems; they continue to operate as long as reactants are supplied. Hydrogen fuel cells offer ultra-clean energy and can have hydrogen shipped to the site if storage is low, increasing reliability. However, they have a relatively high cost and low round-trip efficiency of about 59 % [18].

The stored hydrogen, equivalent to stored energy, is calculated as [22]:

$$E(t)_{stor} = E(t-1)_{stor} + (\Delta E(t)_{grid} / Eff_{inv}) Eff_{elect} \quad (6)$$

where $E(t)_{stor}$ and $E(t-1)_{stor}$ are the energy stored in the hydrogen tanks at hours t and $t-1$, respectively, $\Delta E(t)_{grid}$ is energy mismatch in the grid during hour t , Eff_{inv} is the efficiency of the inverter, and Eff_{elect} is the efficiency of the cell electrolyzer.

This comprehensive overview provides insights into the technical background and operational principles of MG components, focusing on wind turbines, PV modules, diesel generators, batteries, and hydrogen fuel cells. Each component's unique characteristics and performance considerations are essential for effective MG planning and operation.

3. Cost analysis

Cost minimization is a primary objective in any planning study. Estimating the cost of an integrated device within a power system requires consideration of various expenses in addition to the capital installation cost. The total cost can be mathematically formulated as follows [22]:

$$AC_T = \sum (C_{CAP} + C_{REP} + C_{O\&M} - C_S) \quad (7)$$

Where: AC_T , C_{cap} , C_{rep} , $C_{o\&m}$ and C_s stand for the total, nominal capital, replacement, the O&M and salvage costs, respectively.

Political and financial factors significantly influence storage installation costs on both regional and national levels based on market data. Research centers such as the National Renewable Energy Laboratory (NREL) and Sandia National Laboratories provide approximate costs based on manufacturing expenses, taking into account regional financial conditions.

For instance, the Fraunhofer Institute provides estimated 2030 costs for diesel generator fuel in the Middle East, specifically Egypt [23]. This study incorporates the latest estimated cost values from research centers, along with available market costs. Data for wind turbines, PV modules, diesel generators, fuel cells, and lead-acid batteries are sourced from [23,24]. Through curve fitting, an approximate average cost formula is derived for the studied components.

Although wind turbines are relatively cheaper than PV modules for the same rated power, the annual profiles of wind speed and solar radiation are crucial factors in the optimal sizing process. While batteries are generally more affordable than fuel cells, fuel cells enhance overall MG reliability since hydrogen tanks can be stored for emergencies. Additionally, fuel cells are ultra-clean devices with nearly zero emissions, making them suitable for MG distribution. Using the estimated cost equations and predefined models for each MG component, five different MG schemes are analyzed and compared for a typical study case.

The cost of diesel generation can be divided into two components. First, fixed cost includes installation, O&M and other related capital costs. Second, running cost expresses consumed traditional fossil fuel that related to different operating conditions. [25] estimates needed fuel consumption in liter/hr for different diesel generators loading levels. Common experience in planning studies is to divide annual load profile hourly. Therefore, fuel cost per hour describes perfectly desired running cost. Various constraints must be considered when sizing a diesel generator. These can be outlined as follows [26]:

- The capacity of the diesel generator should be sufficient to cover the maximum load and be capable of direct connection to the load when possible.
- If the diesel generator functions as a battery charger, the generated current is restricted by the battery's capacity.
- Diesel generators perform most efficiently when operating between 80 and 90 % of their rated power. Efficiency significantly decreases during light loading conditions.
- Approximately 30 % of the rated power fuel consumption occurs at no load. Therefore, it is advisable to avoid operating below 30 % of full load for extended periods to prevent serious maintenance issues such as chemical corrosion and glazing.

By using a proper curve fitting technique and available costs, quadratic equivalent equations can be concluded for both fixed and running diesel generating cost, as shown in Fig. 1. Table 1 summarizes the factors of MG's different components cost equivalents.

4. Two-stage Lagrange optimization procedure lop

The Lagrange algorithm is employed to determine the optimal

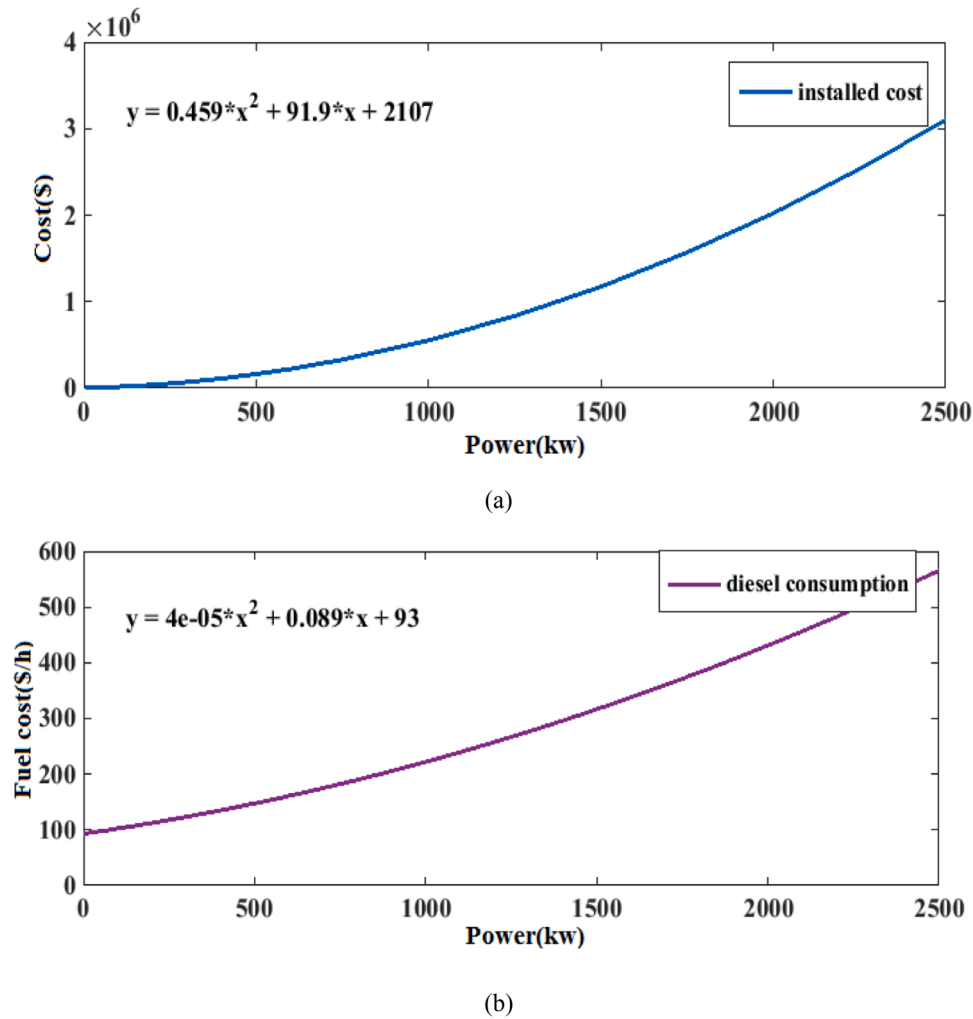


Fig. 1. Equivalent cost equations of diesel generators, a-installed cost b-fuel cost.

Table 1
Equivalent cost equations factors.

Component	a	b	c
PV	-0.088	1300	4000
Wind	0.29	1000	2900
Diesel Generator	0.046	92	2200
Battery	0.00005	0.97	-15
Fuel Cell	-0.0029	5.8	850

solution that minimizes the System Total Investment Cost (STIC), subject to the constraints of the hybrid system components. The primary goal of the proposed algorithm is to economically dispatch power every hour based on the prospective annual demand profile, available natural resources, and predefined planning cost equations.

Stage 1: Generation Sizing Using Lagrange Optimization

The Lagrange multiplier method is mathematically formulated for power system economic dispatch studies as follows [27]:

Minimize: $\sum_{i=1}^n C_i(P_i)$ **subject to** $\sum_{i=1}^n P_i = L$ (8)

Lagrangian function:

$$l(P_1, P_2, \dots, P_n, \lambda) = \sum_{i=1}^n C_i(P_i) + \lambda \left(L - \sum_{i=1}^n P_i \right) \quad (9)$$

Where: P_i represents the production of generating unit i , $C_i(P_i)$ the cost of producing this amount of power with this unit, L is the load demand and λ is a new variable called a Lagrange multiplier.

Setting partial derivatives of this Lagrangian function to zero gives the necessary conditions for optimality. Cost analysis is needed for each studied MG's component to define equivalent cost equations needed for this optimization technique. The proposed algorithm LOP is flexible and easy in coding for any studied generation combinations. With proper costing analysis, it can size both traditional and renewable sources based on available natural resource characteristics and load profile features, as shown in Fig. 2.

In a hybrid PV/wind/diesel microgrid, hourly load power is sourced from different combinations of resources based on the availability of natural resources. If solar radiation (Srad) exceeds a minimum threshold (Smin) and wind speed (Wspd) is above the minimum required level (Wmin), the Lagrange optimization technique allocates demand power according to the cost equations for PV, wind, and diesel generator fuel.

Solar radiation patterns, which vary daily and seasonally, typically begin and end with low radiation levels, leading to negligible power production from PV modules. To prevent the oversizing of PV modules, a minimum solar radiation threshold (Smin) is set at 10 percent of the maximum annual solar radiation at the studied locations.

The hourly optimal profile for PV, wind, and diesel-generated power is determined based on the initial generation conditions. Using accurate models for PV modules and wind turbines, as described in the relevant chapter, hourly sizes are calculated considering wind speed and solar

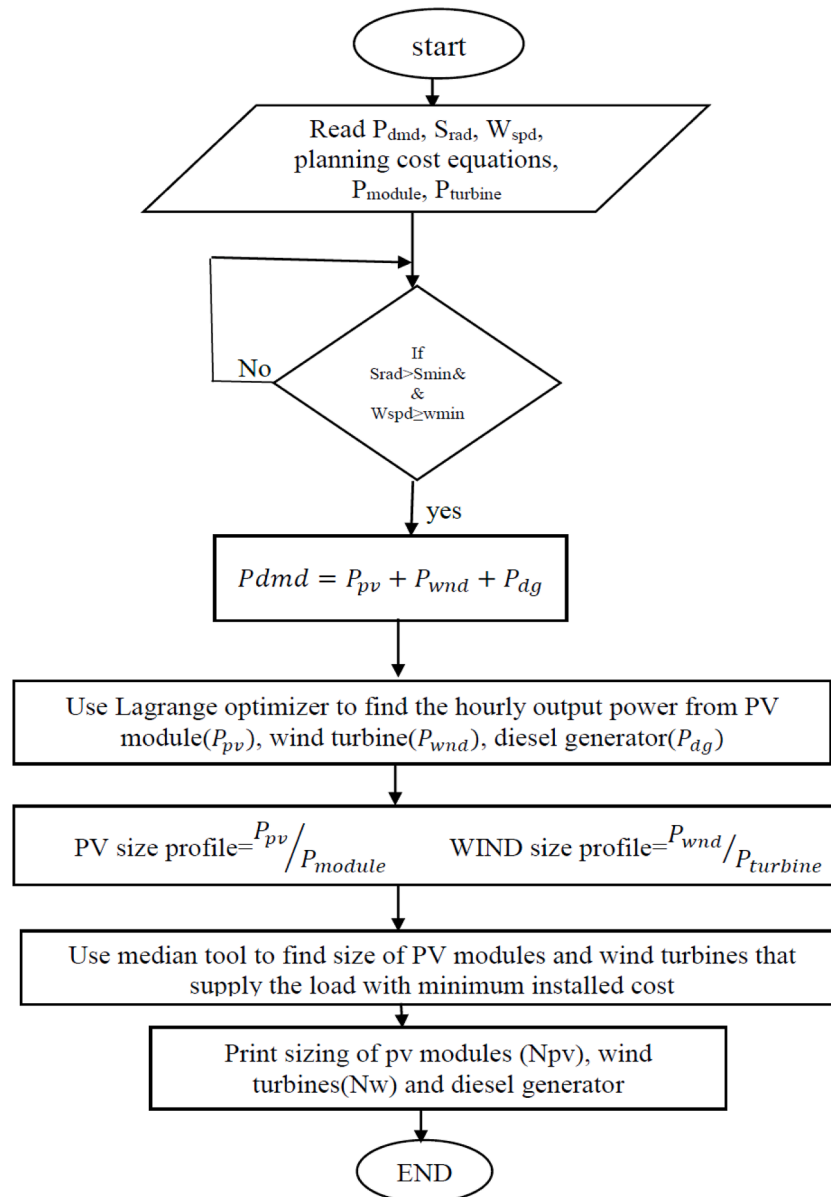


Fig. 2. Stage 1 flowchart.

radiation profiles. A statistical median tool is employed to identify the most suitable sizes for the studied combination, ensuring that the load is met at the minimum planning cost.

Consequently, the optimal sizes of wind turbines and PV modules are estimated. Total generated power is calculated after the generation sizing process, and any excess generated power informs the storage sizing process. A simple energy management algorithm is proposed to validate the planning technique, as discussed in the following subsection.

Stage 2: Energy Storage Sizing and Management

Energy storage systems are indispensable in microgrids (MGs) to manage and mitigate fluctuations between generated power and loads. Batteries are particularly suitable for MG applications due to their adequate capacity and straightforward installation requirements, unlike other technologies such as pumped hydro or compressed air systems discussed earlier in this chapter.

This research focuses on the utilization of batteries within hybrid PV, wind, and diesel generation isolated MGs. Batteries discharge energy

based on their available charge state within specified constraints of State of Charge (SOC). Diesel generators also contribute to storage charging using excess generated power, thereby enhancing overall system reliability. Following the generation sizing process, the total hourly renewable energy output can be forecasted based on local natural sources. Diesel generator usage is minimized during operation to reduce running costs and environmental impact.

The sizing of energy storage depends on the annual difference between generated energy and load requirements, detailed further in subsequent sections. An operational strategy is proposed to address discrepancies between pre-sized generating units and demand load. This strategy involves monitoring the power difference ΔP between MG generation and load, and subsequently deciding whether to charge or discharge the storage under various operational constraints.

Operational limits for batteries and diesel generators are established, forming the basis of the strategy that coordinates between storage and diesel generators:

- During periods of power balance ($\Delta P = 0$), storage remains unchanged with no charging or discharging. Diesel generators do not operate as chargers.
- If ΔP is positive, surplus power charges the storage until it reaches maximum capacity. Diesel generators are inactive as chargers during this phase.
- In periods of generation shortfall, storage and diesel generators compensate for ΔP :
 - Ø If stored energy is within operational limits, storage discharges to meet ΔP , reducing stored energy.
 - Ø Diesel generators activate if stored energy falls below the minimum acceptable limit.
 - Ø Diesel generators operate above their minimum output level ($P_{dg_{min}}$). If required power cannot be met by existing sources, additional diesel-generated power charges the storage, increasing stored energy.

The available stored energy for batteries is calculated using equations outlined in the previous section. Fig. 3 illustrates a flowchart detailing the proposed operational strategy for battery storage systems.

This approach ensures efficient management of MGs by effectively utilizing batteries and diesel generators in response to varying energy generation and demand, contributing to enhanced reliability and sustainability of the microgrid system. The whole algorithm pseudo code is summarized in Fig. 4.

5. Results and discussion

A stand-alone hybrid microgrid (MG) was sized for the Gulf of Al-Aqaba, Egypt (Latitude 27°56'N, Longitude 34°35'E). Meteorological data were taken from wind and solar Atlas Meteorological Service [28–30]. Hourly average wind speed and solar radiation data over a one-year period are analyzed, see Fig. 5. Moreover, Average hourly load demand data is studied, as shown in Fig. 6.

Five different MG configurations were sized and analyzed using the proposed sizing algorithm:

1. PV/battery/diesel system.
2. Wind/battery/diesel system.
3. Wind/PV/battery/diesel system.

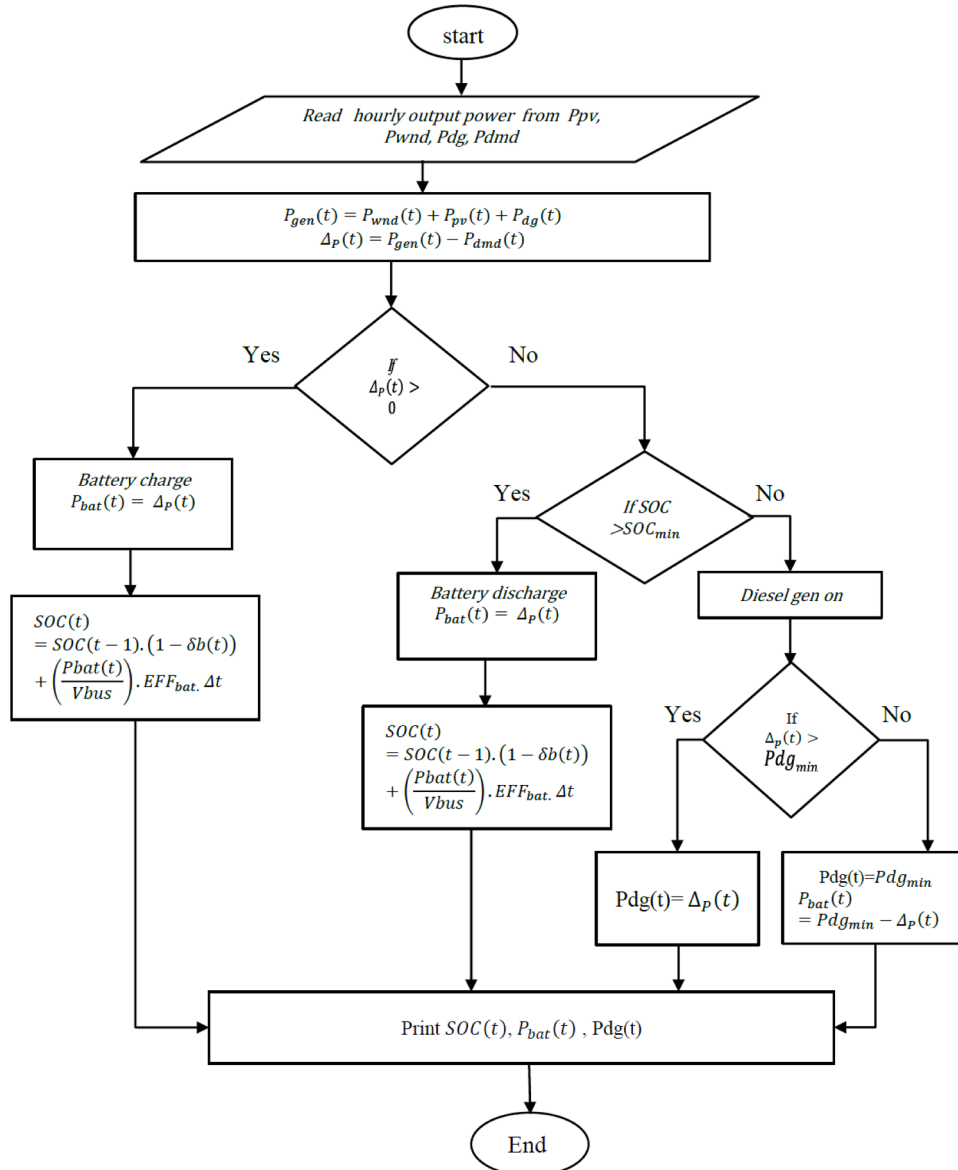


Fig. 3. Stage 2 Flowchart.

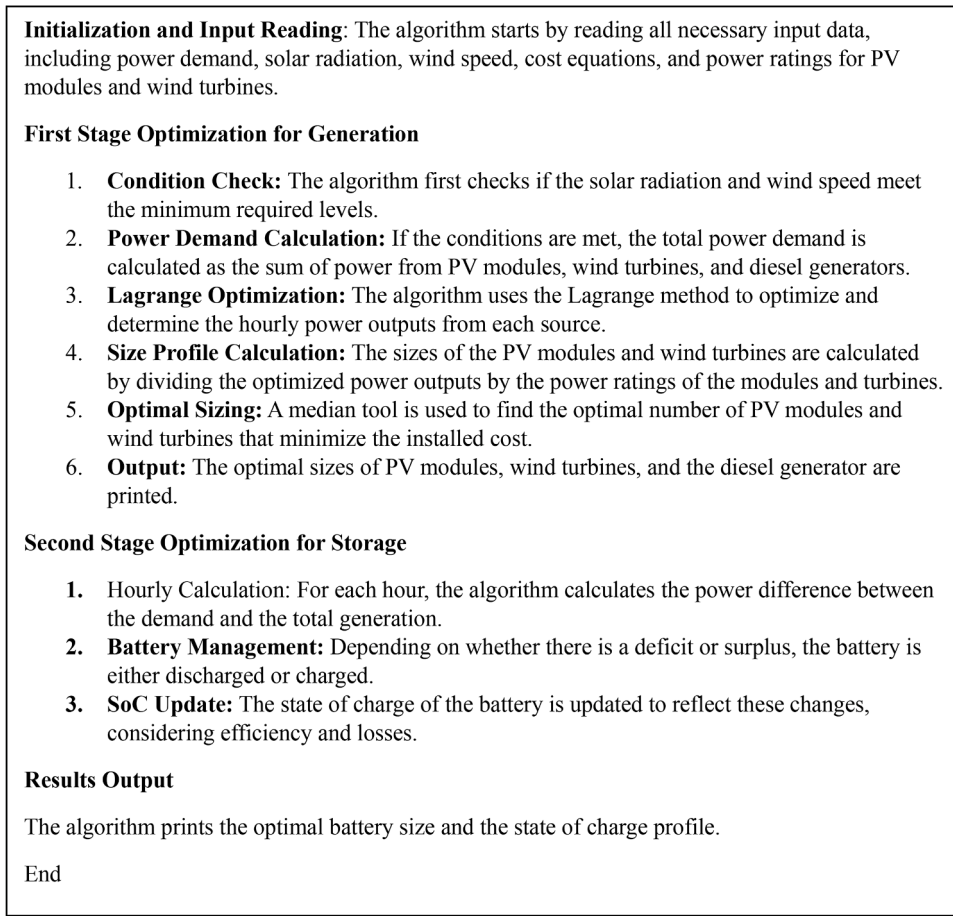


Fig. 4. The proposed algorithm.

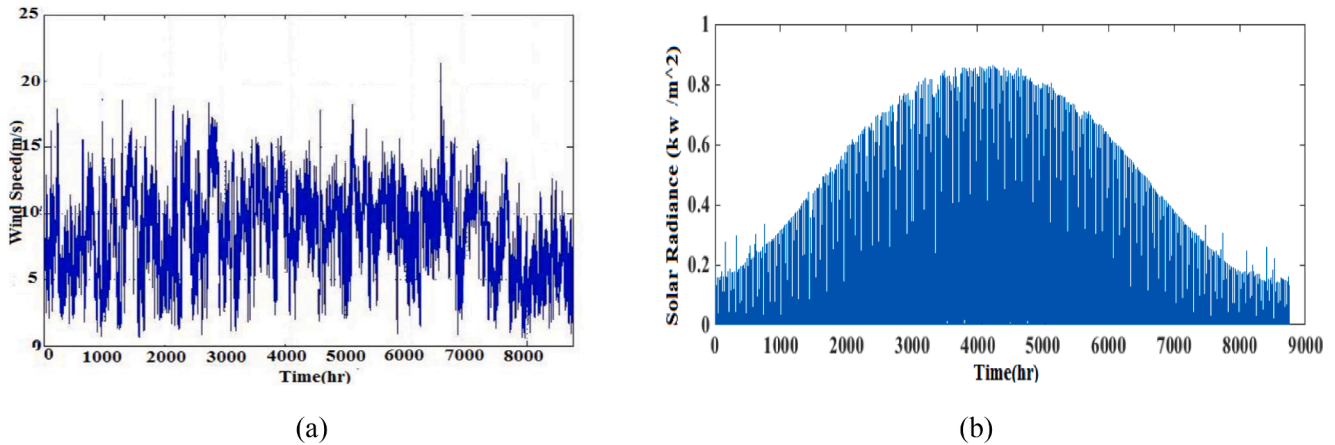


Fig. 5. The studied Meteorological data, a-wind speed, b- global solar radiation.

4. Wind/PV/fuel cell/diesel system.
5. Wind/PV/fuel cell system.

For LOP validation, The MG schemes were re-sized using the HOMER program. Table 2 shows the sizing comparison.

The results show close alignment between LOP and HOMER software, validating the proposed algorithm's effectiveness for different MG schemes. Four criteria were selected to evaluate each scheme:

- **Cost of Electricity (COE):** Indicates the economic benefits of the proposed schemes. COE is defined as the average total cost per kWh of consumed energy by the demand load.
- **Environmental Impact:** Calculated using the Hybrid Optimization Model for Energy Renewables (HOMER), which quantifies the amount of harmful gas emissions produced annually by each scheme.
- **Excess Energy:** Refers to surplus electrical energy that must be dumped because it cannot be used to serve a load or charge the batteries. Excess energy serves as an indicator of scheme reliability.

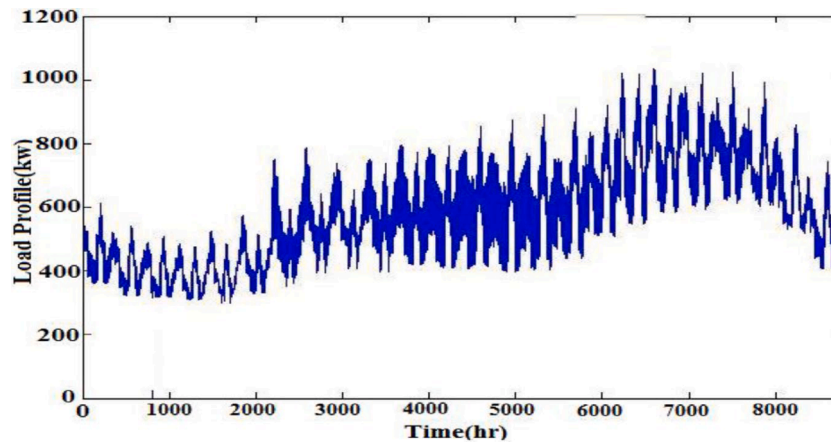


Fig. 6. The studied load profile.

Table 2

LOP and HOMER comparison for different MG schemes.

Scheme	PV Modules (300 kW) - HOMER	PV Modules (300 kW) - LOP	Wind Turbines (330kW) - HOMER	Wind Turbines (330kW) - LOP
S1	4	5	–	–
S2	–	–	10	10
S3	2	2	7	7
S4	1	2	7	7
S5	3	2	11	10

- **Renewable Sharing:** The proportion of renewable energy in each scheme is calculated as an additional comparison factor.

Five MG schemes were designed for the same location characteristics. A comparison was implemented based on four different features: economic benefits, environmental effect, reliability, and renewable energy sharing.

5.1. Economic benefits

Maximizing economic benefits is the main goal of any planning process. These benefits can be measured as the total Net Present Cost (NPC) and Cost of Electricity (COE). NPC is the present value of all the costs incurred over a system's lifetime, minus the present value of all the revenue earned. COE is calculated as the average cost per kWh of useful electrical energy produced. Table 3 shows these criteria for all studied schemes.

Scheme 3 is the most economical, utilizing all available renewable resources (solar and wind) with cost-effective battery storage. Schemes using diesel generators for fuel cell charging are economically inefficient due to the low efficiency of fuel cells.

5.2. Environmental impact

Minimizing emissions is crucial for MGs, especially when located near local loads. Carbon dioxide, carbon monoxide, and nitrogen oxides

Table 3

Economic conditions for different MG schemes.

Scheme	NPC (M\$) - LOP	NPC (M\$) - HOMER	COE (\$/kWh) - LOP	COE (\$/kWh) - HOMER
S1	39.5	44.4	0.30	0.348
S2	23.9	23.3	0.187	0.185
S3	21.8	23.1	0.17	0.183
S4	31.01	29.9	0.243	0.234
S5	29.5	29.7	0.231	0.242

are considered the main harmful emissions from electrical grids. Using the HOMER program, emissions were estimated for each studied scheme, as shown in Table 4.

Scheme 5 has negligible emissions compared to the other MG schemes due to the absence of a diesel generator and battery storage.

5.3. Reliability

Reliability is confirmed by the adequacy of generated power to cover demand loads under various operating conditions. Excess available energy improves MG reliability by increasing the possibility of storing extra energy. Table 5 shows excess available energy for the studied cases.

Scheme 3 provides the highest excess electricity, indicating improved reliability with comprehensive use of renewable resources and efficient storage.

5.4. Renewable sharing

Increasing renewable energy penetration in electrical grids is a global trend to mitigate global warming. Table 6 shows the sharing of renewable energy in each studied scheme.

LOP provides significant cost benefits over HOMER in the PV/Battery/Diesel system. The LOP estimates for NPC are 11.02 % lower than those of HOMER, indicating considerable savings in total system costs. Additionally, COE calculated by LOP is 13.79 % lower, highlighting its potential to reduce operational expenses. Environmentally, LOP estimates carbon dioxide emissions to be 7.16 % lower, suggesting better environmental performance compared to HOMER.

In the Wind/Battery/Diesel system, LOP demonstrates a balanced approach to cost-effectiveness. Although the NPC is slightly higher by 2.58 %, the COE is 1.08 % lower, showing that LOP can maintain economic efficiency. Additionally, LOP calculates 7.72 % lower carbon monoxide emissions, indicating better environmental benefits and a potential reduction in harmful pollutants.

For the Wind/PV/Battery/Diesel system, LOP shows clear economic advantages with an NPC that is 5.63 % lower than HOMER's and a COE that is 7.10 % lower, emphasizing both initial and operational cost savings. Moreover, LOP calculates a 1.76 % higher excess energy, demonstrating improved reliability. The renewable energy contribution is also 4.55 % higher in LOP results, enhancing the system's sustainability and utilization of renewable resources.

The Wind/PV/Fuel Cell/Diesel system benefits from LOP's lower cost estimates, with an NPC that is 3.71 % lower than HOMER's, indicating a reduction in overall costs. The COE is 4.27 % lower, further providing economic advantages. Environmentally, LOP estimates for nitrogen oxides emissions are 9.73 % lower, highlighting significant

Table 4

Environmental emissions for different MG schemes.

Emissions	S1 LOP	S1 HOMER	S2 LOP	S2 HOMER	S3 LOP	S3 HOMER	S4 LOP	S4 HOMER	S5 LOP	S5 HOMER
CO ₂ (10 ³ kg/yr)	3049	3284	616.7	572.2	620.4	761.1	921.6	900.4	Neg.	Neg.
CO (kg/yr)	7527	8106	1522	1413	1532	1879	2275	2145	30.6	30.6
Nitrogen Oxides (kg/yr)	67,164	72,334	13,585	12,604	13,666	16,764	20,30	20,115	136	136

Table 5

Excess electricity for different MG schemes.

Scheme	Excess Energy (kWh/y) - LOP	Excess Energy (kWh/y) - HOMER
S1	88,805	41,521
S2	978,200	1843,000
S3	2541,000	2319,000
S4	2374,000	2572,000
S5	346,957	320,556

Table 6

Renewable and diesel generator sharing of load demand energy.

Scheme	Renewable Share (%) - LOP	Renewable Share (%) - HOMER	Diesel Generator Share (%) - LOP	Diesel Generator Share (%) - HOMER
S1	60	49	40	51
S2	76	85	24	15
S3	92	88	8	12
S4	91	92	9	8
S5	100	100	0	0

environmental improvements and reduced harmful emissions.

In the Wind/PV/Fuel Cell system, LOP offers slight cost benefits with an NPC that is 0.67 % lower than HOMER's. The COE is 4.55 % lower, showing a reduction in operational costs. Environmentally, LOP results show significantly lower emissions, with carbon dioxide and nitrogen oxides emissions 9.66 % and 10.61 % lower, respectively. Additionally, renewable energy contribution is 5.41 % higher in LOP calculations, supporting greater sustainability and less reliance on non-renewable sources.

6. Conclusion

This study presents a robust framework for the optimal sizing and planning of stand-alone microgrids using the Lagrange multiplier optimization technique. Our analysis demonstrates that this method, with its mature and proven effectiveness in power system economic dispatch, significantly reduces system costs and carbon emissions while ensuring reliable and sustainable energy solutions for rural, off-grid communities. The comparative assessment with HOMER software underscores the advantages of the Lagrange multiplier technique, especially in contexts with limited access to advanced computational resources and expertise.

The detailed cost analysis of various hybrid combinations of PV modules, wind turbines, batteries, fuel cells, and diesel generators reveals that the Lagrange method provides superior economic benefits and environmental impact. This makes it an ideal approach for developing countries and low-income rural areas where resource optimization is crucial. By leveraging this technique, we can achieve substantial reductions in greenhouse gas emissions, contributing to global climate change mitigation efforts. Future research could incorporate scalability studies across different geographical locations with varying resource availability to validate the robustness of the proposed method.

CRedit authorship contribution statement

Rasha Elazab: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmed Mamdouh Ewais:** Software, Investigation, Data curation. **Maged Ahmed Abu-Adma:** Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] M. Manas, S. Sharma, K.S. Reddy, et al., A critical review on techno-economic analysis of hybrid renewable energy resources-based microgrids, *J. Eng. Appl. Sci.* 70 (2023) 148, <https://doi.org/10.1186/s44147-023-00290-w>.
- [2] J. Chen, Optimize configuration of multi-energy storage system in a standalone microgrid, *Front. Energy Res.* 11 (2023) 1283859, <https://doi.org/10.3389/fenrg.2023.1283859>.
- [3] M. Abidin, D. Said, A review on the microgrid sizing and performance optimization by metaheuristic algorithms for energy management strategies, *E3S Web Conf.* 516 (2024) 01008, <https://doi.org/10.1051/e3sconf/202451601008>.
- [4] A.S. Tukkee, N.I.B.A. Wahab, N.F.B. Mailah, Optimal sizing of autonomous hybrid microgrids with economic analysis using grey wolf optimizer technique. *e-Prime - Adv. Electr. Eng. Electron. Energy* 3 (2023) 100123.
- [5] M. Kharrich, S. Kamel, R. Ellaia, Economic and ecological design of hybrid renewable energy systems based on a developed IWO/BSA algorithm, *Electron* 10 (2021) 687.
- [6] A. Abdelkader, A. Rabeh, D.M. Ali, J. Mohamed, Multi-objective genetic algorithm-based sizing optimization of a stand-alone wind/PV power supply system with enhanced battery/supercapacitor hybrid energy storage, *Energy* 163 (2018) 351.
- [7] J. Aguila-Leon, C. Vargas-Salgado, C. Chinas-Palacios, D. Diaz-Bello, Energy management model for a standalone hybrid microgrid through a particle swarm optimization and artificial neural networks approach, *Energy Convers. Manag.* 267 (2022) 115920.
- [8] C.A. Nallolla, V.P.D. Chittathuru, S. Padmanaban, Multi-Objective optimization algorithms for a hybrid AC/DC microgrid using res: a comprehensive review, *Electron* 12 (2023) 1062.
- [9] A. Baakeer, M.F. Elmorshedy, H. Salama, et al., Optimal design and performance analysis of coastal microgrid using different optimization algorithms, *Electr. Eng.* 105 (2023) 4499–4523, <https://doi.org/10.1007/s00202-023-01954-9>.
- [10] L.F. Grisales-Noreña, J.A. Ocampo-Toro, O.D. Montoya-Giraldo, et al., Optimal operation of battery storage systems in standalone and grid-connected DC microgrids using parallel metaheuristic optimization algorithms, *J. Energy Storage* 65 (2023) 107240.
- [11] A. Elnozahy, H.S. Ramadan, F.K. Abo-Elyousr, Efficient metaheuristic Utopia-based multiobjective solutions of optimal battery-mix storage for microgrids, *J. Clean. Prod.* 303 (2021) 127038.
- [12] M.F. Roslan, M.A. Hannan, P.J. Ker, R.A. Begum, T.M. Indra Mahlia, Z.Y. Dong, Scheduling controller for microgrids energy management system using optimization algorithm in achieving cost saving and emission reduction, *Appl. Energy* 292 (2021) 116883.
- [13] P.P. Kumar, R.P. Saini, Optimization of an offgrid integrated hybrid renewable energy system with different battery technologies for rural electrification in India, *J. Energy Storage* 32 (2020) 101912.
- [14] A. Fathy, K. Kaaniche, T.M. Alanazi, Recent approach based social spider optimizer for optimal sizing of hybrid PV/Wind/Battery/Diesel integrated microgrid in aljouf region, *IEEE Access* 8 (2020) 57630.
- [15] N. Alamir, S. Kamel, T.F. Megahed, M. Hori, S.M. Abdelkader, Developing hybrid demand response technique for energy management in microgrid based on pelican optimization algorithm, *Electric Power Syst. Res.* 214 (2023) 108905.
- [16] İ. Çetinbaş, B. Tamyürek, M. Demirtaş, Sizing optimization and design of an autonomous AC microgrid for commercial loads using Harris Hawks Optimization algorithm, *Energy Convers. Manag.* 245 (2021) 114562.

- [17] A.L. Bukar, C.W. Tan, L.K. Yiew, R. Ayop, W.S. Tan, A rule-based energy management scheme for long-term optimal capacity planning of grid-independent microgrid optimized by multi-objective grasshopper optimization algorithm, *Energy Convers. Manag.* 221 (2020) 113161.
- [18] A.M. Ewais, R. El-Azab, M.A.A. Adma, Stand-alone microgrid energy storage schemes: comparative study, in: *Proceedings of the 2018 Twentieth International Middle East Power Systems Conference (MEPCON)*, Cairo, Egypt, 2018, pp. 564–569, <https://doi.org/10.1109/MEPCON.2018.8635266>.
- [19] J.K. Kaldellis, An integrated model for performance simulation of hybrid wind-diesel systems, *Renew. Energy* 32 (2007) 1544–1564.
- [20] A.E.S.A. Nafeh, proposed technique for optimally sizing a PV/diesel hybrid system, in: *Proceedings of the 696 International Conference on Renewable Energies and Power Quality*, Spain, 2010, 23–25 March.
- [21] EPRI-DOE Handbook of Energy Storage For Transmission & Distribution Applications, EPRI; U.S. Department of Energy, Palo Alto, CA; Washington D.C., 2003.
- [22] Y. Kalinci, A. Hepbasli, I. Dincer, Techno-economic analysis of a stand-alone hybrid renewable energy system with hydrogen production and storage options, *Int. J. Hydrog. Energy* (2014).
- [23] N. saad Hussein, M. Abokersh, C.K.H. Schlegel, "Electricity cost from renewable energy technologies In Egypt" FRAUNHOFER Institute For Solar Energy Systems ISE Version: 2016.
- [24] Renewables and Electricity Storage A technology Roadmap For REmap 2030, The International Renewable Energy Agency (IRENA), 2015.
- [25] Approximate fuel consumption chart is available on, www.dieselserviceandsupply.com. 2024.
- [26] V. Khare, S. Nema, P. Baredar, Optimization of the hybrid renewable energy system by HOMER, PSO and CPSO for the study area, *Int. J. Sustain. Energy* (2015).
- [27] P. Pedregal, Introduction. In: *introduction to optimization*. Texts in Applied Mathematics, Springer, New York, NY, 2004, https://doi.org/10.1007/0-387-21680-4_1 vol 46.
- [28] S. Pfenninger nger, I. Staffell, Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data, *Energy* 114 (2016) 1251–1265, <https://doi.org/10.1016/j.energy.2016.08.060>.
- [29] I. Staffell, S. Pfenninger, Using bias-corrected reanalysis to simulate current and future wind power output, *Energy* 114 (2016) 1224–1239, <https://doi.org/10.1016/j.energy.2016.08.068>.
- [30] The datasets are available in the [Renewable. Ninja] repository, 2024 [<https://www.renewables.ninja/>], Accessed on 20/6/2024.